User guide

A practical guide to the current native vocal ensemble effect.

What I.R.Choir does

I.R.Choir creates an ensemble from a vocal audio input using delayed, modulated and pitch-shifted layers. You can position and mix those layers, shape their synthetic tone colors, and add master effects.

> Prototype boundary: the result is not a trained choir conversion model. Bass, tenor, alto and soprano labels organize processed parts; they are not sampled human voice types.

The signal path

Vocal input → choir layers and part processing → dry/wet blend → low/high EQ → saturation → linked compression → delay → algorithmic room → output gain → optional mono audition.

Choose a simple starting workflow

Goal	Starting point
Subtle vocal thickening	Tight Doubles; low Room and moderate Choir Mix.
Wide backing layer	Wide Backing or Wide; adjust Stereo Width and monitor mono.
Atmospheric texture	Cloud or Dream Haze; listen carefully to consonants and pitch stability.
Relative harmony	Choose four relative parts and edit the four interval knobs.
MIDI harmony	Route input audio plus MIDI notes; use a clear monophonic guide.

Before important work

Test on a copy of a DAW project. Listen at matched loudness against bypass. The automated tests establish software behavior, not musical realism or target-host reliability.

The interactive stage

Action	Result
Click a singer	Select it for individual level, pan, mute, solo and synthetic color.
Shift-click	Add or remove singers from the selection.
Drag a selected singer	Move the selection in stereo and adjust its level. Vertical movement is visually limited to keep rows legible.
Click empty stage space	Clear a group selection before selecting one singer.
Double-click	Toggle that singer mute.
Bass / Tenor / Alto / Soprano	Select every active layer in that part.
All singers	Select every active layer.
Group pan	Apply a relative pan offset; the control returns to zero after dragging.
Group level	Set all selected layers to the displayed level.
Group mute / solo	Toggle the selection together. Solo affects wet layers only.

Formations

Semicircle curves the stage; Tight cluster narrows placement; Wide rows spreads singers across the stereo field. Applying a formation resets singer pan and level. Reapply after changing ensemble size to space the current singers again.

Animation

Full and Subtle modes animate the actual layer meters. Distinct faces, hairstyles and choir robes identify the visual layers; these are not separate voice models. Reduced motion removes mouth and singer glow animation while retaining slow meter refresh. The Humanize sliders provide a conventional alternative to dragging the conductor pad.

Performance and humanize

Control	What changes
Ensemble Size	2 to 32 active layers. Higher counts increase processing cost.
Choir Mix	Blend the original vocal and processed ensemble. At 100%, only wet layers remain.
Stereo Width	Scale individual pan positions; there is no pan effect in a mono output.
Human Motion	Slow delay modulation across singers.
Timing Looseness	Vary the small delay offsets between layers.
Pitch Drift	Slow pitch variation, expressed in cents.
Vibrato	Additional pitch movement in cents.
Intensity	Layer gain and modulation intensity; the conductor vertical axis.
Space	Controls ambience strength with Room; the conductor horizontal axis.
Synthetic Character	Blend each layer with its warm/airy/dark/bright filter color.
Output	Final gain after all effects. Clip indication is not limiting.

Getting a natural result

Use small timing and pitch amounts before increasing layer count. Keep some dry signal for intelligibility. Excessive width or movement can weaken consonants or produce phase cancellation. A 12-layer setting is not automatically better than a 4-layer setting.

Synthetic voice colors

Warm, Airy, Dark and Bright are tonal filters, not voice identities. Distinct recorded or AI singers require separately licensed recordings or a trained model that is not included.

Effects and metering

Effect	Use and limits
Low / High EQ	Tone shelves around the low and high regions; not a full surgical equalizer.
Saturation	Adds nonlinear color. Increase carefully and compare at matched loudness.
Compression	Linked stereo envelope control to reduce high-level peaks; no automatic makeup gain.
Delay Mix	Add delayed repeats.
Delay Time	50-1000 ms. Changes are smoothed and can create tape-like pitch movement.
Feedback	Controls repeat decay; limited to 0.8 for stability.
Room type	Six algorithmic spaces, including Chamber, Plate-style and Tunnel-style; no measured impulse responses.
Room	Master ambience blend; also affected by the conductor Space control.

Read the meters

L/R show peak output in dBFS. CLIP stays lit after an overload until cleared. Correlation approaches +1 for similar left/right signals and can go negative when they oppose each other. Mono RMS measures the level after summing, before Mono audition is applied.

Use Mono audition while adjusting width. If the choir becomes much quieter or thin, reduce width and motion and compare again. The vocal guide frequency is an approximate pitch estimate, not a tuning meter.

Smoothing

The engine ramps major mix, level, gate, conductor and FX changes. This reduces abrupt control jumps but does not establish artifact-free sound under every automation pattern.

Shape the reverb

Open Reverb space from the control-page menu.

Choose Studio, Hall, Cathedral, Chamber, Plate-style or Tunnel-style. All are algorithmic settings; no EastWest recordings or convolution impulse responses are included.

Control	Action
Reverb on	Enable the wet return; switching off restores the dry signal with a short fade.
Mono send	Sum the input sent to the reverb while leaving the dry stereo image intact.
Low cut / High cut	Toggle the send filters; frequency knobs set their cutoff.
Freeze	Hold the reverb tail. Turn off to release it; lower Room before experimenting.
Wet only	Hear the reverb return without the dry vocal. Reverb off restores dry audio.
Pre-delay	0-200 ms before the reverb; Tunnel-style adds 9 ms of spacing.
Decay shape	Relative room decay adjustment, not calibrated seconds.
Reverb width	Set the stereo width of the wet return.
Damping	Soften high frequencies as the reverb decays.
Room / conductor Space	Control the wet return amount from Performance and the conductor pad.

Start with Hall, a low Room level, 12-30 ms pre-delay and some low cut. Use wet-only listening to tune the space, then return to the full mix. These recommendations are starting points for listening, not measured acoustic models.

Presets, favourites and history

Factory preset	Intended character
Intimate	4 restrained layers with little ambience.
Wide	12 broad backing layers.
Cloud	24 layers with more motion and space.
Massive	32 dense layers.
Tight Doubles	2 layers, subtle movement, no room.
Wide Backing	12 layers, a little compression and high tone.
Dream Haze	24 layers with delay and darker high tone.
Cathedral	16 layers with the largest algorithmic room.
Crowd Wash	32 layers with stronger movement and color.

Save and find your own sounds

Type a name and click Save. A new .irchoir.xml file is created in your user application-data location under IRChoir/Presets. Existing files are not silently overwritten. Star the current preset to mark it as a favourite; Favourites filters the browser.

A/B and undo

A/B saves the current side before loading the other. Copy sends the current sound to the other side. Both snapshots survive closing the editor while the plugin instance exists, but only the current sound is saved with a DAW project.

Undo and Redo cover parameter edits. Whole preset-file loading and A/B replacement are separate state operations. Preset names do not show every unsaved change. Loading a factory preset resets sonic parameters and singer edits; animation preference is retained.

Four parts and MIDI

Relative parts

Each active layer is assigned to one of four repeating parts. Set their levels independently and choose intervals from -12 to +12 semitones. The default offsets are -12, -5, 0 and +7 relative to your incoming vocal. This creates processed harmony layers; it does not change the source into four authentic human voice types.

MIDI targets

Choose MIDI note targets, route MIDI into the plugin using your host, and send a stable monophonic vocal to the audio input. The engine estimates the input pitch and shifts parts toward held MIDI notes. Sustain and channel-aware note release are supported.

Behavior	Current implementation
Which notes?	The lowest four unique held notes. When fewer notes are held, the highest available note repeats.
Shift range	At most one octave up or down relative to the detected input.
No held notes	The wet layers are gated off. Dry signal may remain, depending on Choir Mix.
No stable pitch	The MIDI wet layers are gated off; unvoiced sounds can be affected.
Guide range	Approximate monophonic detection, about 65-1000 Hz.
Formants	No formant preservation; large shifts can sound synthetic.

Example

With a clear A3 guide (about 220 Hz), try held notes A2, E3, A3 and E4. Begin with low movement and no master effects. Judge note transitions and consonant clarity before adding ambience.

AU routing

The AU is a MIDI-controlled audio effect. In Logic and other hosts, the route may differ from a normal audio insert. MIDI alone does not create sound; a vocal audio source is still required.

Troubleshooting and care

Symptom	Check
No plugin in the DAW	Confirm a real Mac build exists, the correct format was installed, and a rescan completed. Read host scan errors.
No wet sound	Check audio input, mutes/solos, Choir Mix and MIDI held notes if MIDI mode is selected.
Metallic or smeared sound	Reduce shifts, drift, vibrato and motion. Try fewer layers and a cleaner vocal. Granular artifacts remain a known limitation.
Clipping	Reduce Output, drive, EQ boost, delay feedback and layer level.
Weak mono sound	Reduce width/motion, audition the mono sum and watch correlation.
Preset cannot save	Check application-data folder permissions. The UI reports failure rather than claiming success.
Too much CPU	Use fewer layers. Try Subtle or Reduced motion. Measure in the actual DAW at your buffer size.
macOS blocks the file	Verify the build origin and signing status. Do not disable Gatekeeper globally.

Remove the plugin

Quit the DAW and remove IRChoir.component from /Library/Audio/Plug-Ins/Components and IRChoir.vst3 from /Library/Audio/Plug-Ins/VST3. Rescan. Your separately stored presets remain.

Report a reproducible problem

Record the plugin version, Mac model/chip, macOS, DAW/version, AU or VST3, sample rate, buffer size, preset and exact steps. Include scan/crash logs. Supply a short audio example only if you own or have permission to share it.

Before public release: Mac validation, installer checks, target-DAW testing and real-vocal listening are still required. Consult the Release and Test Report for the exact executed coverage.